

Silicon Valley Bank

A Risk Analysis Case Study

Group 66

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Comparison Bank: The Bank of New York Mellon
Scenario F

1. Executive Summary

As of December 31, 2022, Silicon Valley Bank (SVB) exhibited a risk profile that, in retrospect, made failure highly predictable. The bank had an exceptionally large long-duration securities portfolio funded almost entirely by short-term, uninsured deposits. The bank's failure was structural balance sheet fragility triggered by an interest rate and liquidity shock.

Interest rate risk was the dominant vulnerability. SVB's one-year cumulative repricing gap stood at negative \$102.5 billion, implying a \$1.0 billion reduction in net interest income (NII) for every 100 basis points of parallel rate increase. Combined unrealized losses on the AFS and HTM securities portfolios totalled an estimated \$17.7 billion, exceeding Tier 1 Capital of \$17.0 billion. A duration-gap analysis under a 200 bps shock yields an estimated equity value loss of \$26.1 billion, more than 1.5 times Tier 1 Capital.

Liquidity risk was also big. Non-retirement deposits above \$250,000 represented 75.0% of total assets: nearly double the peer average of 45.9%. Under Scenario F, SVB's capital falls from well-capitalised to undercapitalised, with a stressed Tier 1 capital ratio of 6.6% and a leverage ratio of 3.5%. Credit risk was benign: the Texas Ratio was 0.88% and RAROC on the commercial portfolio was 36.7%, far above its 3.35% benchmark.

2. Interest Rate Risk

2.1 Repricing Gap

On the asset side, only 35.7% of total assets repriced within one year, well below the peer average of 42.9%, while 51.9% were long-term (greater than five years) - compared to just 32.2% for peers. On the liability side, 84.7% of liabilities repriced within one year versus 79.7% for peers. This produces a severe mismatch: a one-year cumulative repricing gap of negative \$102.5 billion (approximately negative 49% of total assets).

Item	RSA \$000s	RSL \$000s	GAP \$000s	Cum. GAP \$000s
Up to 1 Year	74,559,574	177,024,119	-102,464,545	-102,464,545
> 1 to 5 Years	20,359,132	0	20,359,132	-82,105,413
> 5 Years	108,526,299	0	108,526,299	26,420,886

NII sensitivity: Under a uniform +100 bps shock, the repricing model estimates a change in NII of negative \$1.02 billion, equivalent to a 22.2% decline relative to baseline NII of \$4.61 billion. Under +200 bps, the estimated NII reduction doubles down to negative \$2.05 billion. Even under an asymmetric scenario (RSA +150 bps, RSL +75 bps), NII still declines by \$209 million, which only confirms that the liability-sensitive position dominates regardless of assumption.

2.2 Unrealized Losses

By year-end 2022, SVB had accumulated estimated unrealized losses of \$2.5 billion on its AFS portfolio and \$15.2 billion on its HTM portfolio, totalling \$17.7 billion. Expressed as a percentage of Tier 1 Capital, the combined unrealized loss ratio was negative 104.1% - placing SVB in the 3rd percentile of its peer group. In contrast, the peer average was negative 32.0%. Our interpretation is that the economic value of SVB's equity had effectively been wiped out, even though the regulatory capital ratios - which exclude HTM mark-to-market losses - showed the bank as well-capitalised.

Item	\$000s	%
AFS Unrealized Loss	-2,525,457	-14.86
HTM Unrealized Loss	-15,159,540	-89.20
Combined AFS+HTM	-17,684,997	-104.06

2.3 Duration Gap Analysis

Weighted-average asset duration of 7.43 years and liability duration of 1.12 years mean the duration gap is 6.39 years - indicating that the market value of assets is far more sensitive to interest rate increases than the market value of liabilities. When applied with a 200 bps parallel shock, the estimated change in equity value is negative \$26.1 billion - approximately 154% of reported Tier 1 Capital of \$17.0 billion.

Why do regulatory ratios and duration-gap analysis give conflicting signals? Regulatory capital ratios are computed using book-value accounting. HTM securities are carried at amortised cost, meaning their unrealised losses are invisible to the regulatory capital framework. Duration-gap analysis on the other hand captures the full mark-to-market impact on economic equity. Hence, the two frameworks show fundamentally different pictures: the former showed SVB as comfortably well-capitalised (CET1 of 15.3%), while the latter revealed that, on a mark-to-market basis, SVB was already economically insolvent before the bank run even started.

3. Liquidity Risk

3.1 Deposit Concentration

Non-retirement deposits above \$250,000 as a share of total assets: SVB's ratio stood at 75.0%, nearly double the peer group average of 45.9%, placing it in the 93rd percentile. This metric is the single most important liquidity vulnerability in SVB's profile. Almost the entire deposit base was uninsured (\$151.6 billion out of \$175.4 billion total deposits, or 86.4%), and most of these depositors were venture-backed start-ups with payroll obligations - a homogeneous, highly correlated cohort that would act simultaneously under stress.

3.2 Liquid Asset Composition

The UBPR reports SVB's Liquid Assets FV at 55.2% of total assets (\$115.4 billion), which superficially looks favourable relative to the peer average of 38.0%. However, this figure is misleading because it includes HTM securities at fair value, which are structurally unavailable in a liquidity stress without realising losses and triggering a reclassification that would expose the entirety of the book to mark-to-market accounting.

Item	\$000s	%
Cash & Due from Banks	1,191,448	0.57
Interest-Bearing Bank Balances	11,329,209	5.42
Fed Funds Sold & Resales	710,688	0.34
Cash-like Assets (subtotal)	13,231,346	6.33
AFS Securities (Fair Value)	25,981,932	12.43
Cash-like + AFS Securities	39,213,278	18.76
HTM Securities (Fair Value)	76,169,074	36.44

Effective liquidity coverage: Using cash-like assets plus AFS securities (\$39.2 billion) as the genuinely usable liquidity buffer, SVB could absorb only 25.9% of its uninsured deposits (\$151.6 billion) before needing to liquidate HTM securities and realise the embedded losses. Given that the actual deposit outflow on March 9-10, 2023 reportedly reached approximately \$42 billion in a single day, this buffer was wholly inadequate. HTM securities, although theoretically convertible, are far less liquid under stress: selling them forces reclassification of the remaining portfolio to AFS, crystallises mark-to-market losses, and further erodes regulatory capital, which creates a self-reinforcing spiral that effectively renders them unusable as a first-resort buffer.

4. Credit Risk

By every conventional metric, SVB's credit quality at the time of the report was benign. Non-current loans + OREO as a share of total loans were just 0.19%, against a peer average of 0.61%. The net charge-off rate was 0.10% versus 0.24% for peers. The ACL coverage ratio of 4.78x non-accrual loans exceeded the peer average of 3.49x.

Texas Ratio: SVB's Texas Ratio was 0.88%. A Texas Ratio above 100% is traditionally associated with elevated failure risk; so the figure we calculated for SVB is negligible, confirming that credit losses posed no near-term solvency threat.

RAROC: The risk-adjusted return on capital for SVB's commercial and industrial loan portfolio was 36.7%, far above the hurdle rate benchmark of 3.35% (cost of funds plus 275 bps). The C&I portfolio was generating strong risk-adjusted returns and did not contribute to the bank's fragility.

Based on this, we conclude that the credit risk was not the driver of SVB's failure. The bank's loan portfolio was well-provisioned, performing, and highly profitable on a risk-adjusted basis. The failure was hence caused entirely by the interaction of interest rate risk and liquidity risk.

5. Stress Scenario Analysis - Scenario F

Assigned parameters: Interest rate shock +100 bps, NPL increase +4.0 percentage points, deposit outflow 45% of uninsured deposits.

5.1 Channel 1: Interest Rate Risk - NII Impact

Item	Data
Cumulative 1-Year GAP (\$000s)	-102,464,545
Rate Shock	+100 bps (0.01)
ΔNII (\$000s)	-1,024,645
Baseline NII (\$000s)	4,611,646
Stressed NII (\$000s)	3,587,001
% Change in NII	-22.2%

Even though the +100 bps shock in Scenario F is the smallest of any rate-up scenario, the liability-sensitive gap means the income impact is still severe: a \$1.0 billion reduction in NII, or more than one-fifth of annual interest income. This can be translated directly into weaker retained earnings and therefore reduced capital-building capacity going forward.

5.2 Channel 2: Credit Risk - Additional Provisions

Item	Data
Baseline NPL Ratio	0.19%
NPL Increase (Scenario F)	+4.0 pp
Stressed NPL Ratio	4.19%
New Non-Performing Loans (\$000s)	2,969,960
LGD Assumption	45%
Additional Provisions Required (\$000s)	1,336,482
Stressed ACL (\$000s)	1,972,482

The 4.0 pp NPL increase is the largest of any scenario, reflecting, in a way, a severe credit stress in SVB's VC-backed loan book. New non-performing loans of \$3.0 billion require additional provisions of \$1.34 billion, more than doubling the existing ACL from \$636 million to \$1.97 billion. These provisions flow directly through the income statement, thereby reducing retained earnings and Tier 1 Capital.

5.3 Channel 3: Liquidity Risk - Deposit Outflow

Item	Data
Uninsured Deposits (\$000s)	151,592,000
Outflow Rate (Scenario F)	45%
Deposits Lost (\$000s)	68,216,400
Cash-like Assets (\$000s)	13,231,346
Shortfall after Cash-like Assets (\$000s)	-54,985,054
AFS Securities Available (\$000s)	25,981,932
Shortfall after AFS Sales (\$000s)	-29,003,122
HTM Securities Sold (\$000s)	29,003,122

A 45% outflow from uninsured deposits triggers a \$68.2 billion liquidity demand that SVB cannot even remotely meet from cash-like assets alone. After exhausting all \$13.2 billion of cash-like assets, a \$55.0 billion shortfall remains. And selling the entire AFS portfolio (\$26.0 billion) still leaves a \$29.0 billion gap, forcing the liquidation of \$29.0 billion of HTM securities - more than a third of the HTM book.

5.4 Channel 4: Capital Adequacy - Stressed Ratios and PCA

Item	Data
Baseline Tier 1 Capital (\$000s)	16,995,000
Less: Additional Provisions (\$000s)	-1,336,482
Realized Loss on AFS Sales (\$000s)	-2,525,457
Realized Loss on HTM Sales (\$000s)	-5,772,343
Total Realized Securities Losses (\$000s)	-8,297,800

Stressed Tier 1 Capital (\$000s)	7,360,718
Total RWA (\$000s)	111,341,000
Stressed Tier 1 Ratio	6.61%
Stressed Leverage Ratio	3.52%
PCA Category under Stress	Undercapitalised

PCA classification: The stressed leverage ratio of 3.52% is below the 4% minimum for adequately capitalised and the 5% minimum for well-capitalised, placing SVB in the undercapitalised PCA category. The stressed Tier 1 ratio of 6.61% is above the 6% adequately-capitalised threshold but below the 8% well-capitalised threshold. The binding constraint is therefore the leverage ratio, which reflects the sheer size of the total asset base relative to the already dramatically reduced Tier 1 Capital.

Our interpretation and conclusion is hence that even if the stressed ratios remained marginally above minimum thresholds, SVB would remain deeply vulnerable in practice. The combination of a \$17.7 billion pre-existing unrealised loss overhang, a fragile depositor base, and a management credibility crisis following the March 2023 capital-raising announcement means that any residual capital adequacy would be immediately questioned by markets and depositors. The PCA framework assesses static end-of-period capital ratios; it cannot capture the dynamic feedback loop between loss realisation, depositor confidence, and further outflows.

6. Peer Comparison - SVB vs The Bank of New York Mellon

Item	SVB	BNYM	Difference
Long-Term Assets to Total Assets (%)	51.92%	19.87%	+32.05 pp
AFS/HTM Unrl. Loss to Tier 1 Cap (%)	-104.06%	-60.65%	-43.41 pp
Non-Retirement Dep >\$250K to Assets (%)	74.99%	56.63%	+18.36 pp
Net Loans & Leases to Deposits (%)	41.97%	11.15%	+30.82 pp
Net Interest Income to Avg. Earn. Assets (%)	2.20%	1.00%	+1.20 pp
CET1 Capital Ratio (%)	15.26%	16.32%	-1.05 pp
Leverage Ratio (%)	7.96%	6.25%	+1.72 pp

Long-term asset concentration: SVB held 51.9% of total assets in instruments maturing beyond five years, compared to just 19.9% at BNY Mellon. This reflects SVB's strategic decision to deploy its 2020-21 deposit inflows into long-duration MBS and Treasuries to boost yield in a low-rate environment. BNY Mellon, as a custody and clearing bank, maintains a far shorter-duration balance sheet, making it substantially less exposed to rate-driven asset-value declines.

Unrealised loss exposure: Both banks suffered large unrealised losses by the end of 2022, but SVB's combined AFS+HTM loss ratio of negative 104.1% of Tier 1 Capital was obviously far worse than BNY Mellon's negative 60.7%. The difference reflects the more extreme duration of SVB's portfolio rather than a categorically different strategy.

Uninsured deposit concentration: SVB's 75.0% ratio of large uninsured deposits to total assets versus 56.6% at BNY Mellon reflects a business model difference: SVB served a narrow, homogeneous tech and VC client base, whereas BNY Mellon's custody business generates more diverse institutional flows. The concentration of SVB's depositors in a single interconnected industry created herd behaviour that caused a same-day bank run.

In conclusion, the Q4 2022 data unambiguously identifies SVB as unusually fragile relative to its peer.

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